

1 Collaborative Filtering

Given the following user-item interaction matrix in a recommender system. Rows denote users and columns denote items.

	$p1$	$p2$	$p3$	$p4$	$p5$	$p6$
$u1$	?	5	?	4	?	3
$u2$	?	2	4	2	4	?
$u3$	3	?	4	?	?	2
$u4$	?	4	1	?	1	?

1. Which user is most similar to $u2$ based on cosine similarity?
2. Which user is most similar to $u2$ by using Pearson correlation coefficient?
3. Predict the rating for $p6$ of user $u4$. Use Pearson Correlation Coefficient for user similarity computation

2 Evaluation of Recommender Systems

Suppose we have two recommendation algorithms A and B. We trained the two algorithms on some dataset, and test them in the dataset as follows in the form of triples (user, item, rating):

$$(u1, p3, 3), (u2, p1, 2), (u2, p4, 1), (u2, p5, 3), (u3, p1, 4), (u3, p3, 5)$$

Let the two algorithms output the following predicted ratings:

Algorithm	User	Predicted ratings $\hat{r}$ for all items not rated in training data
A	$u1$	$p3 : 2.5, p4 : 3$
	$u2$	$p1 : 3, p4 : 2, p5 : 2.5$
	$u3$	$p1 : 2, p3 : 3, p5 : 1$
B	$u1$	$p3 : 3.5, p4 : 3$
	$u2$	$p1 : 2, p4 : 3, p5 : 4$
	$u3$	$p1 : 4, p3 : 3, p5 : 2$

1. Compute the MSE for both algorithms.

2. Consider the top-N recommendation problem and convert the ground truth data into binary labels (like or dislike) in the test data as follows:

(1) Any rating above (including) 3 stars denote that the user like the item;

(2) Any missing value or rating below 3 is considered as dislike. Calculate the Precision@1 and Recall@1 for the two algorithms.

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Given the following user-item interaction matrix in a recommender system. Rows denote users and columns denote items.

Pearson correlation coefficient:

	p1	p2	p3	p4	p5	p6
u1	?	5	?	4	?	3
u2	?	2	4	2	4	?
u3	3	?	4	?	?	2
u4	?	1	1	?	1	?

1. Apply the basic user-based collaborative filtering with Pearson correlation coefficient for user u4 to predict the rating for p6.

2. Extend the above user-based CF with bias. Predict the rating for p6 of user u4.

$$W_{ik} = \frac{\sum_{j \in S_{ik}} (r_{ij} - \bar{r}_i)(r_{kj} - \bar{r}_k)}{\sqrt{\sum_{j \in S_{ik}} (r_{ij} - \bar{r}_i)^2} \cdot \sqrt{\sum_{j \in S_{ik}} (r_{kj} - \bar{r}_k)^2}}$$

$$1. \text{Sim}(u_1, u_4) = \frac{(5-4) \cdot (4-2)}{\sqrt{(5-4)^2} \cdot \sqrt{(4-2)^2}} = 1$$

$$\bar{u}_1 = \frac{5+4+3}{3} = 4 \quad \bar{u}_4 = \frac{4+1+1}{3} = 2$$

$$\text{Sim}(u_3, u_4) = \frac{(4-3) \cdot (1-2)}{\sqrt{(4-3)^2} \cdot \sqrt{(1-2)^2}} = -1$$

$$\bar{u}_3 = \frac{3+4+2}{3} = 3$$

$$\hat{r}(u_4, p_6) = \frac{3 \cdot 1 + 2 \cdot (-1)}{|1| + |1|}$$

$$W_{ik} = \frac{\sum_j s_{ijk} (r_{ij} - \bar{r}_i) (r_{kj} - \bar{r}_k)}{\sqrt{\sum_j s_{ijk} (r_{ij} - \bar{r}_i)^2} \cdot \sqrt{\sum_j s_{ijk} (r_{kj} - \bar{r}_k)^2}}$$

$$1. \quad \text{sim}(u_1, u_4) = \frac{(5-4) \cdot (4-2)}{\sqrt{(5-4)^2} \cdot \sqrt{(4-2)^2}} = 1$$

$$\bar{u}_1 = \frac{5+4+3}{3} = 4 \quad \bar{u}_4 = \frac{4+1+1}{3} = 2$$

$$\text{sim}(u_3, u_4) = \frac{(4-3) \cdot (1-2)}{\sqrt{(4-3)^2} \cdot \sqrt{(1-2)^2}} = -1$$

$$\bar{u}_3 = \frac{3+4+2}{3} = 3$$

$$\hat{r}(u_4, p_6) = \frac{3 \cdot 1 + 2 \cdot (-1)}{|1| + |-1|} = 0.5$$

$$2. \quad b_g = \frac{39}{13} = 3.$$

b_{u1} = average score of u_1 - b_g

$$= 4 - 3 = 1$$

$$b_{u1, p_6} = b_g + b_{u1} + b_{p_6}$$

$$= 3.5$$

b_{u3} = average score of u_3 - b_g

$$= 3 - 3 = 0.$$

$$b_{u3, p_6} = 3 - 0.5 = 2.5$$

$$b_{u4, p_6} = 2 - 0.5 = 1.5$$

b_{p_6} = average score of p_6 - b_g

$$b_{u1, p_6} = 2.5 - 3 = -0.5$$

$$\hat{r}(u_4, p_6) = \frac{1.5 + (3-3.5) \cdot 1 + (2-2.5) \cdot (-1)}{|1| + |-1|}$$

b_{u4, p_6}

$|1| + |-1|$

b_{u3, p_6}

1. MSE: $\frac{1}{N_{\text{test}}} \sum_{r_{ij} \in \text{testset}} (r_{ij} - \hat{r}_{ij})^2$. MSE(A) = 1.75, MSE(B) = 1.54. Algorithm B is better.

Algorithm	User	Top-1 Item	Groundtruth	# tp	# fp	# fn
A	u1	p4	p3	<u>0</u>	1	<u>1</u>
	u2	p1	p5	<u>0</u>	1	<u>1</u>
	u3	p3	p1, p3	<u>1</u>	0	<u>1</u>
B	u1	p3	p3	1	<u>0</u>	0
	u2	p5	p5	1	0	0
	u3	p1	p1, p3	1	0	1

2. The top-1 recommendation for the two algorithms is listed in the table above.

Algorithm A, Precision@1 is $\frac{1}{3}(\underline{0/1} + \underline{0/1} + \underline{1/1}) = \frac{1}{3}$, Recall@1 is $\frac{1}{3}(0/1 + 0/1 + 1/2) = \frac{1}{6}$.

Algorithm B, Precision@1 is $\frac{1}{3}(1/1 + 1/1 + 1/1) = 1$, Recall@1 is $\frac{1}{3}(1/1 + 1/1 + 1/2) = 2.5/3 \approx 0.83$.

3 Latent Factor Model

ROC

+

AUC